

Duc Pham

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EDUCATION

Purdue University – West Lafayette, IN

M.S + B.S in Computer Engineering

Expected May 2027

GPA: 3.6/4.0

- Relevant Coursework:** Advanced Computer Architecture, System-on-Chip Design, Hardware & Software Security, ASIC Design, Operating Systems, Artificial Intelligence, Embedded Systems, Digital Systems Design, Data Structures & Algorithms

EXPERIENCE

Tesla, Inc. – AI Hardware Intern

Next-gen AI6 Inference SoC

Jan 2026 – Jun 2026

Palo Alto, CA

- Formally verified an AI6 MSHR-backed **L1 Cache coherence** system in JasperGold using SVA, proving MESI liveness, snoop-filter consistency, and multi-core atomic ordering invariants across full L1 property set pre-silicon.
- Verified an out-of-order systolic MAC array via a sequential equivalence checking (SEC) RTL-to-C strategy; mapped pipeline scoreboards and drove COI/proof-core coverage closure to trap microarchitectural hazards and denormal/NaN cases.
- Architected formal abstractions to overcome state-space explosion on 100K-FLOP compute unit, leveraging k-induction, cut-points, black-boxing, and symbolic variables for full proof convergence.
- Extended an async Pytest validation framework for AP DUTs with dependency-injected fixtures, FIO/stress-ng workloads; added plugins for target discovery, Splunk telemetry, and CI board provisioning.

EPFL Processor Architecture Lab – Research Intern

High-Performance Task Parallelism FPGA & HBM System Characterization

May 2025 – Sep 2025

Lausanne, Switzerland

- Built a Chisel-based task-level-parallel (TLP) architecture on AMD Alveo U280, achieving **128x** speedup across 128 PEs at 12% LUT overhead; removed memory bottlenecks for **8%** gain using a custom cycle-accurate simulator.
- Designed an HBM traffic generator via a pipelined AXI5 DMA controller in Chisel with DecoupledIO interfaces and store-and-forward buffering, sustaining a throughput of 12.8 Gb/s at 200 MHz.
- Closed **85%** functional coverage on the TLP fabric and AXI5 DMA datapath via a cocotb regression suite with constrained-random traffic and a golden reference model, targeting arbitration, burst-boundary, and back-pressure cases.

SoCET AI Hardware – Architecture Developer

Tensor Core's Scratchpad for GEMM/Conv Acceleration

Jan 2024 – Present

West Lafayette, IN

- Integrated shared memory subsystem with vector core & systolic array at 900 MHz via 9-stage pipelined Benes-network permutation crossbar, featuring 8-way SRAM banking, swizzling, load/store coalescing.
- Built a UVM testbench using constrained-random traffic and scoreboard checking, driving functional coverage closure on bank-conflict and permutation cases; achieved **1.2x** GEMM speedup, **10%** higher DRAM utilization, and **15%** lower L2 traffic.

PROJECTS

NAND SSD FTL with ML-Guided Policy Search (Agentic FTL) – NextSys Lab

May 2026 – Present

- Designed the garbage-collection and data-placement policies in a discrete-event NAND SSD performance model, cutting write amplification **31%** and P99 tail latency **59%** versus a greedy baseline under skewed writes.
- Drove design-space exploration and workload characterization in C++ across steady-state, equal-write-volume sweeps over over-provisioning, skew, and wear leveling, plus performance-model validation.
- Led architectural trade-off analysis showing greedy victim selection was already near-optimal, redirecting the roadmap to hot/cold data placement and ML-guided policy search tuned for WAF, endurance, and QoS latency.

Dual-Core Pipelined RISC-V Processor with Coherent L1 Cache

Aug 2025 – Dec 2025

- Developed a dual-core, 5-stage pipelined RISC-V processor with coherent L1 caches and a custom control unit supporting parallel execution; implemented multithreading & synchronization with RISC-V assembly validation.
- Closed timing at 82 MHz on AMD Spartan-7 FPGA (Urbana board) with a pipelined bus, delivering **2x** higher throughput over the single-core baseline while demonstrating cache-coherence protocol correctness across cores.

TECHNICAL SKILLS

Languages & Tools: SystemVerilog/Verilog, C/C++, gem5, Python, Chisel/Scala, Perl, TCL, Bash, RISC-V Assembly, PrimeTime, Vivado, Quartus, Verdi, QuestaSim, VCS, Verilator, VC Formal, JasperGold, Jenkins, cocotb, Virtuoso, Palladium, MATLAB, Git, Linux
Design & Architecture: Microarchitecture (5-stage pipelines, out-of-order execution, branch prediction, crossbars), cache coherence (MESI, snoop filters, MSHR), memory subsystem (multi-bank SRAM, HBM, DDR, prefetching), memory consistency, NoC, AMBA AXI4/AXI5, RISC-V ISA, CDC, synthesis (Design Compiler, Genus), DPI-C, static timing analysis, timing closure, FPGA prototyping
Methodologies: dynamic verification (UVM, constrained-random, scoreboard checking, functional coverage closure), formal verification (SVA authoring, k-induction, COI/proof-core coverage), pre-silicon validation (regression, waveform debug, CI/CD)